Previous Year Paper 2012

1. Which of the following is/are cited by the scientists as evidence/evidences for the continued expansion of universe?
   1. Detection of microwaves in space.
   2. Observation of redshift phenomenon in space.
   3. Movement of asteroids in space.
   4. Occurrence of supernova explosions in space.

   Select the correct answer using the codes given below:
   (a) 1 and 2
   (b) 2 only
   (c) 1,3,4
   (d) None of the above
2. Electrically charged particles from space travelling at speed of several hundred km/sec can severely harm living beings if they reach the surface of the earth. What prevents them from reaching the surface of the Earth?
   (a) The earth's magnetic field diverts them towards its poles.
   (b) Ozone layer around the Earth reflects them back to outer space.
   (c) Moisture in the upper layers of atmosphere prevents them from reaching the surface of the Earth.
   (d) None of the above are correct.
3. Government of India encourages the cultivation of 'sea buckthorn'. What is the importance of this plant?
   1. It helps in controlling soil erosion and in preventing desertification.
   2. It is a rich source of biodiesel.
   3. It has nutritional value and is well-adapted to live in cold areas of high altitudes.
   4. Its timber is of great commercial value.

   Which of the statements given above is /are correct?